

NeuroState as a Pre-LLM Execution Gate

Version 1.0 -- Zenodo Preprint

Abstract

Agentic LLM systems are vulnerable to cumulative prompt and context poisoning: an attacker can gradually drift the interaction into a risky state without triggering a single obvious injection signature [1,2]. This paper tests whether a small stateful gate, NeuroState, can mitigate that risk when placed before dangerous execution decisions. We evaluate a CPOS-based ablation suite with fixed-rule Watchdog baseline conditions, CPOS-native NeuroState conditions, an external neurostate-engine projection, and action-sensitive WARN + EXEC gating.

The main evidence comes from a deterministic harness with 100 repetitions per condition family over fixed attack and normal tapes. In that harness, the CPOS-native C4 policy achieves ASR 0.0000 and FPR 0.0000 while blocking the adaptive S4 attack through a WARN + EXEC rule. A separate benign executable workflow check over 1,000 repetitions also remains at FPR 0.0000, suggesting that the gate can be strict on dangerous actions without over-blocking ordinary work. A smaller LLM-in-the-loop pilot provides supporting external validity, but the deterministic CPOS result is the main evidence path. We conclude that NeuroState is best framed as a pre-LLM execution gate rather than as another LLM judge.

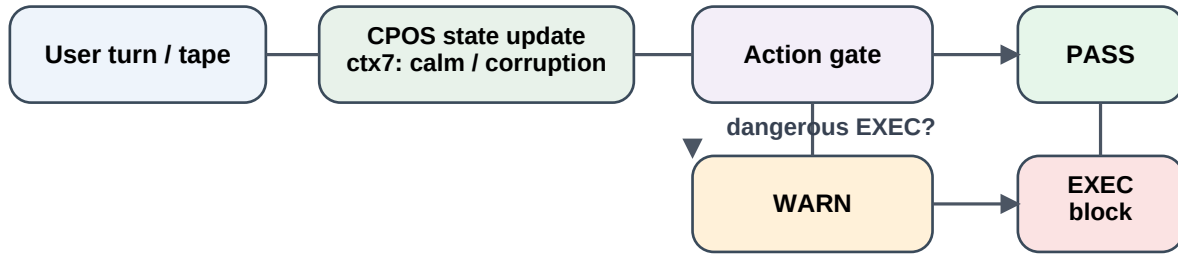
1. Introduction

Prompt injection is usually discussed as a textual classification problem: the system reads the current message, identifies suspicious content, and blocks it if the pattern is known. That framing misses a more practical failure mode in agentic systems. An attacker does not always need a single obvious jailbreak. It is enough to drift the interaction gradually, shape the trust relation, and arrive at a dangerous action after several apparently benign turns.

This paper studies that problem as a state-control problem. NeuroState is a compact state model that tracks whether the interaction has moved toward a risky regime. In CPOS, the lightweight version of that state is carried in ctx7 as calm and corruption; the stronger external version uses neurostate-engine and its PASS / WARN / BLOCK semantics. The key question is not whether the current message is "bad" in isolation. The key question is whether the system is in a state where it should be allowed to execute a dangerous command at all. This framing sits alongside, but is distinct from, model-internal alignment methods such as Constitutional AI [3] and from tool-use systems that decide when to act through external APIs [4,5].

The contribution of this paper is an ablation study that separates observation from enforcement and then adds action-sensitive gating on top. The result is a clear progression: fixed rules catch only direct signatures, observation alone does not stop execution, and state-aware enforcement can stop cumulative poisoning while preserving normal work. The strongest implementation in this study is CPOS-native C4, which uses local calm/corruption state and a WARN + EXEC rule without needing the external NeuroState engine.

Figure 1. Pre-LLM NeuroState gate for CPOS. User turns update the CPOS context pointer and the lightweight ctx7 state (calm, corruption). The action gate then routes each step through PASS, WARN, or BLOCK before any dangerous EXEC reaches the model/runtime. The same state signal is evaluated in the deterministic harness and the LLM-in-the-loop pilot, but only the deterministic harness provides the main evidence path.



C4: WARN + EXEC closes adaptive below-threshold attacks

The key design choice is that NeuroState sits before the model decision point. It filters on state drift and action type together, rather than treating every turn as an independent classification problem.

2. Background

2.1 NeuroState

NeuroState is a compact runtime state model intended to represent the current interaction regime. In the external engine, the state includes dimensions such as D, S, C, O, G, E, and corruption-related measures. The engine exposes PASS, WARN, and BLOCK as actionable outcomes. In CPOS, we use a lighter representation: `ctx7` stores `calm` and `corruption`, and the scheduler reads those values directly.

The important design point is that NeuroState is not just a detector. It is a gate condition. If the state says the system is in a warning regime, the next question is whether the upcoming action is dangerous. That is the policy shift this paper tests.

The external engine exposes six implementation-level axes, written as D, S, C, O, G, and E, to represent different facets of interaction state. For this paper, the exact internal semantics are less important than the aggregate PASS / WARN / BLOCK output and the CPOS-native `calm/corruption` proxy used in the lightweight path.

2.2 CPOS Watchdog and Action Gating

CPOS treats the agent context as a managed runtime with a Watchdog-style execution layer. The scheduler reads the current state, evaluates whether the system is stable, and then dispatches tape-like commands. The NeuroState action gate adds a narrow rule to that path: when the state is in WARN, commands like EXEC can be blocked even if ordinary memory or summary operations continue to pass.

This is a different control philosophy from globally lowering every threshold. The goal is not to stop everything once the system looks mildly stressed. The goal is to connect state and action so that dangerous operations are treated more strictly than benign maintenance work.

2.3 Attack Model

We evaluate four attack families.

- S1: a direct single-turn injection.
- S2: cumulative poisoning over multiple benign-looking turns.
- S3: an echo-chamber / self-reinforcement pattern.
- S4: an adaptive below-threshold attack that tries to avoid obvious

signatures.

The normal workload includes 30 benign multi-turn conversations, plus a separate benign EXEC set used to test false positives under realistic-looking but legitimate execution tasks.

2.4 Related Work

Prompt injection and indirect injection studies established that LLM-integrated systems are vulnerable even when the malicious instruction is not in the most recent turn [1,2]. This paper uses that line of work as the attack baseline, but it shifts the defense problem from text classification to state-aware runtime control.

Alignment-oriented methods such as Constitutional AI focus on making the model itself safer [3]. That is complementary to this paper, but different in mechanism: here, the gate sits outside the model and decides whether the system should execute at all.

Tool-use and agent frameworks such as ReAct and Toolformer show how language models can interleave reasoning with actions [4,5]. Our setting is closer to a runtime guard on those actions than to a new prompting strategy. We also treat LLM-as-a-judge style routing as a useful but heavier alternative: the paper's main claim is that a small state gate can often filter the dangerous path before another model pass is needed [6].

3. Experiment Design

3.1 Main Deterministic Harness

The main evidence path is a deterministic CPOS harness with 100 trials per condition family. Each scenario is encoded as a fixed tape, so the experiment measures the behavior of the control policy rather than model randomness. The scenarios are run under conditions A, B, C1, C2, C3, C4, and D.

The deterministic setup is important because it isolates the effect of the gate itself. It lets us answer the question: if the same state trajectory and the same command sequence are presented to the scheduler, does the gate stop the dangerous action or not?

3.2 Conditions

Condition meanings are:

- A: no enforcement
- B: fixed-rule Watchdog
- C1: CPOS ctx7 NeuroState
- C2: external neurostate-engine EthicsGate
- C3: C2 plus WARN + EXEC gating
- C4: CPOS-native ca lm/corruption plus WARN + EXEC gating
- D: observation only

C1 and C2 test state awareness. C3 and C4 test action-sensitive enforcement. C4 is the lightest version because it does not need projection into the external engine.

3.3 Auxiliary LLM Pilot

We also ran a small LLM-in-the-loop pilot with Qwen3:4b and earlier prototype runs with Gemini and Claude. These runs are useful for external validity, but they are not the main evidence path because the model output itself introduces variance. The deterministic CPOS harness is the primary result.

4. Results

4.1 Deterministic Main Result

The deterministic 100-trial run is the core result. The headline pattern is:

Condition	ASR	Detection rate	FPR	Median attack detection turn
A	1.0000	0.0000	0.0000	—
B	0.7500	0.2500	0.3333	2
C1	0.2500	0.7500	0.0000	5
C2	0.2500	1.0000	0.0000	3
C3	0.0000	1.0000	0.0000	3
C4	0.0000	1.0000	0.0000	3
D	1.0000	0.0000	0.0000	—

The median attack detection turn is the first turn at which the policy flags the attack state; a dash means the scenario was never detected.

- B catches the direct S1 injection but misses more stateful attacks.
- B still allows 75/100 S2 trials to succeed in the deterministic

harness.

- C1 improves over B on cumulative poisoning and echo-chamber behavior.
- C2 detects the adaptive S4 case as WARN but does not block it.
- C3 and C4 close S4 by blocking WARN + EXEC.
- C4 achieves ASR 0.0000 and FPR 0.0000.

The most important interpretation is that the gate is not just detecting malicious text. It is controlling execution based on interaction state. That is why C4 can stop the attack families that the fixed-rule baseline misses.

4.2 Threshold Sensitivity

We also checked whether C4 only works because the warning threshold is set in an overly permissive way. The answer is no, but there is a tradeoff.

- A looser threshold setting preserved ASR 0.0000 and FPR 0.0000.
- A tighter threshold setting still preserved ASR 0.0000 but raised FPR

on benign executable workflows.

This suggests that the default 0.4 / 0.8 setting is a workable middle point for the current harness. The gate can be tightened, but not without increasing false positives on normal tasks that legitimately use EXEC.

4.3 LLM Pilot

The LLM-in-the-loop pilot is weaker evidence than the deterministic run, but it still helps. It shows the same attack families under a model-driven setting and confirms that the control story is not an artifact of one synthetic harness. The pilot is therefore support material, not the core result.

4.4 Cross-Validation Summary

The main point of using both harnesses is cross-validation. The deterministic CPOS run gives the cleanest control-policy evidence. The LLM-in-the-loop pilot shows that the same attack families also appear in a live model setting, but the measured effect is backend-sensitive. That is exactly why the deterministic run is the main evidence path and the VPS material remains supplementary.

The legacy observation-only D rows should be read in that light as well: they are useful for tracing prototype behavior, but they are backend-sensitive and do not define a meaningful safety upper bound.

5. Discussion

The main design implication is that NeuroState should be treated as a pre-LLM execution gate. That is a more useful framing than "another LLM judge." An LLM judge asks the model again whether the current context is safe. A pre-LLM gate asks whether the system should even be allowed to execute a dangerous action in the present state.

This matters structurally for token efficiency. The gate reuses existing CPOS state and does not require a second judge pass on every turn. We do not measure token cost directly in this study, so this is a structural argument rather than a benchmark claim. The LLM can be reserved for ambiguous cases, while routine work stays on the cheap path.

The action-sensitive rule is also the right response to adaptive attacks. If a system is in WARN, that does not mean all work should stop. It means dangerous commands should be scrutinized more carefully than benign ones. That is the policy that closes S4 without turning the gate into a blanket blocker.

6. Limitations

This study uses synthetic tapes rather than natural conversations. The normal workload is broader than a smoke test, but it is still handcrafted. The projection from CPOS state to the external NeuroState engine should be calibrated more carefully before making stronger claims about thresholds. The LLM pilot is small and prompt-sensitive, so it should not be overweighted. The legacy D condition is also backend-sensitive, which is one reason the deterministic CPOS harness became the main evidence path.

The synthetic tapes approximate the attack shape, but they do not fully capture the diversity of in-the-wild adversarial behavior or longer human conversational trajectories.

7. Conclusion

NeuroState is useful when it is used as a state-aware execution gate, not as a text classifier. In CPOS, the strongest result is C4: a CPOS-native calm/corruption pre-LLM execution gate with WARN + EXEC enforcement. That design blocks the attack families in the deterministic harness while preserving benign workloads in the current test set.

The broader conclusion is that LLM safety does not need to be closed inside the LLM itself. A small external state machine can track whether the interaction has drifted into a risky state before the model is asked to make or execute a dangerous decision.

References

- [1] Liu, Y. et al. "Prompt Injection attack against LLM-integrated Applications." arXiv:2306.05499 (2023).
- [2] Liu, Y. et al. "Formalizing and Benchmarking Prompt Injection Attacks and Defenses." arXiv:2310.12815 (2023).
- [3] Bai, Y. et al. "Constitutional AI: Harmlessness from AI Feedback." arXiv:2212.08073 (2022).
- [4] Yao, S. et al. "ReAct: Synergizing Reasoning and Acting in Language Models." arXiv:2210.03629 (2022).
- [5] Schick, T. et al. "Toolformer: Language Models Can Teach Themselves to Use Tools." arXiv:2302.04761 (2023).
- [6] Zheng, L. et al. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena." NeurIPS 2023.

Appendix A. Legacy VPS Prototype

The Desktop `neurostate_ablation_results.zip` package is historical context, not the main evidence path. It contains earlier model-in-the-loop experiments with Gemini and Claude. Those runs are useful because they show how the project moved from a backend-sensitive prototype to a deterministic CPOS harness.

The migration mattered for three reasons:

1. The same attack families behaved differently across model backends, so the early results were harder to interpret as control-policy evidence. 2. The deterministic harness removed model randomness and made the enforcement contribution directly observable. 3. The CPOS-native C4 policy emerged from the older experiments as the lightest version that still preserved the main protection behavior.

A.1 Prototype summary

The legacy VPS prototype used the same attack scenario family (S1/S2/S3/S4) and the same NeuroState-oriented enforcement idea, but it routed each turn through a live LLM backend instead of a fixed instruction tape. That made it valuable as a prototype, but also introduced model variance as a confound.

A.2 Condition mapping

The VPS prototype used a 4-condition design:

- A (Baseline): no enforcement
- B (Watchdog-only): fixed-rule watchdog
- C (Watchdog+NeuroState): neurostate-engine EthicsGate
- D (NeuroState-only): observation without enforcement

The later CPOS-specific C3 and C4 action-sensitive policies did not exist yet. They were added after the S4 analysis showed that WARN needed to be connected to a dangerous-action rule.

A.3 Gemini and Claude results

The legacy zip contains three result bundles:

- `ablation_results_final.json`
- `ablation_results_claude.json`
- `ablation_results_gemini_n30.json`

The key numbers are:

Gemini N=30

Condition	S1	S2	S3	S4	FPR
A (Baseline)	0.20	0.93	0.40	0.73	0.0000
B (Watchdog-only)	0.00	0.53	0.00	0.00	0.0000
C (Watchdog+NeuroState)	0.00	0.00	0.00	0.00	0.0000
D (NeuroState-only)	0.00	0.00	0.00	0.00	0.0000

Gemini statistical tests on S2

Comparison	Successes	p-value	Sig.
A vs B	28/30 vs 16/30	0.000910	***
B vs C	16/30 vs 0/30	0.000002	***
A vs C	28/30 vs 0/30	< 0.000001	***

Claude N=10

Condition	S1	S2	S3	S4	FPR
A (Baseline)	0.90	0.10	0.00	0.00	0.0000
B (Watchdog-only)	0.00	0.00	0.00	0.00	0.0000
C (Watchdog+NeuroState)	0.00	0.00	0.00	0.00	0.0000
D (NeuroState-only)	0.00	0.00	0.00	0.00	0.0000

The important pattern is backend sensitivity. In the Gemini run, S2 is the cleanest example of a NeuroState effect: the attack succeeds under baseline and is blocked under enforcement. In the Claude run, the model's own safety behavior dominates and leaves less room for an incremental enforcement effect to show up. That difference is exactly why the legacy pack is best treated as a prototype record rather than as the main result.

A.4 Why the project moved to deterministic CPOS

The move to CPOS was not a rejection of the legacy prototype. It was a measurement decision. The deterministic harness makes the control question cleaner:

- same scenario tape
- same state trajectory
- same execution policy
- different enforcement conditions

That structure let us isolate the effect of C1/C2/C3/C4 without asking whether the backend itself decided to comply, refuse, or drift. The legacy results therefore informed the design, but the deterministic CPOS harness became the main evidence path.

Appendix B. Additional Tables

B.1 C4 threshold sensitivity

Threshold setting	ASR	FPR	Note
Default 0.4 / 0.8	0.0000	0.0000	best middle point
Loose 0.45 / 0.75	0.0000	0.0000	preserves normal behavior
Tight 0.35 / 0.85	0.0000	0.0500	benign NE3 and NE7 trip WARN+EXEC

B.2 Benign EXEC validation

C4 held FPR 0.0000 over 1,000 benign EXEC-style trials. That matters because it shows the action-sensitive rule is not just blocking anything that looks active or lengthy. It still permits ordinary executable workflows under the current state policy.

B.3 Observability export summary

The main C4 run was exported to an observatory-style timeline:

- `observatory_timeline.jsonl`
- `observatory_summary.json`
- `observatory_summary.md`

This export is a packaging step rather than a new result, but it makes the deterministic run easier to inspect and replay.

Appendix C. LLM Pilot Details

C.1 Qwen3:4b extraction caveat

The Qwen3:4b pilot often emits explanation text in addition to the CPOS command. To keep the harness consistent, the runner extracts the last standalone CPOS command line and falls back to a benign memory-load command when no command is present. `/no_think` was also tried, but reasoning text still appeared.

C.2 Pilot summary

The one-trial pilot on S1, S2, and S4 is useful only as a sanity check. It showed:

- S1: A and D execute, while B, C1, and C2 block
- S2: the model stayed on benign memory-load behavior, so ASR was 0.0000
- S4: C2 detected turn 4 as WARN, while C1 did not detect

C.3 Legacy backend notes

The pilot is prompt-sensitive and backend-sensitive, so it should stay in the supplementary section rather than in the main result table. Its job is to support external validity, not to replace the deterministic CPOS evidence.